

Assignment 1- Spiker Box Experiment (video)

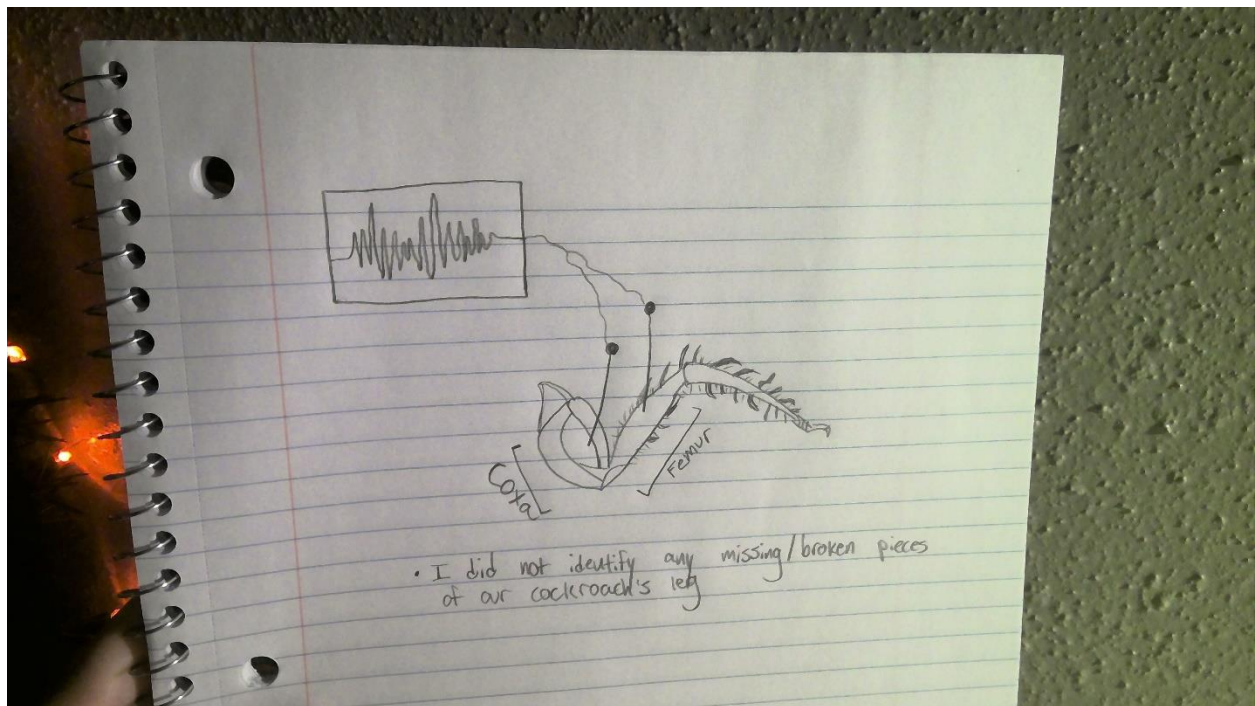
Purpose: Gain hands-on experience a common lab practice of recording neuronal activity via electrophysiology. Though recording precise data of neuronal firing requires thousands of dollars' worth of equipment, this assignment will allow you to witness general trends in neuronal firing that you have only read about in your previous coursework.

Discussion Questions:

1. What method of recording is he using to hear the spikes (i.e., intracellular or extracellular)? **Extracellular**
2. Is he recording a single neuron or multiple neurons? How do we know this? **He is recording multiple neurons (extracellular signals), and we know this because the needle/electrode used is too big to impale and measure a single neuron. Also, we can see many peaks/waves in the data reading because these are all different action potentials, each for a different neuron.**
3. Why does lowering the temperature of the cockroach make it stop moving? What are other ways you could anaesthetize it? **The cockroach becomes anaesthetized when lowered into cold water because of their physiology. Due to the fact that cockroaches are cold-blooded (like the majority of insect/reptiles), placing them in an even colder environment causes their body temperature to lower until it matches that of the water (and thus they lose control of body and cellular processes). Other methods of anaesthetizing it include the following (to name a few): water deprivation, oxygen deprivation (Carbon Dioxide gas exposure), isoflurane (anesthesia), and isopropyl alcohol immersion**
4. If you continuously stimulate one barb on the leg of the cockroach, would spiking activity change? Would spiking activity continue to be elicited by every instance of stimulation, would it eventually attenuate, or would it increase? Why does this occur and how might this process be useful? **Yes and no. The spikes will remain constant for a while if the stimulus/touch is the same every time, but eventually the spikes will reduce in their amplitude until they attenuate (completely stop). This, referred to as "sensory attenuation," occurs because the information being encoded is the same every time (there is no new/modified information and so it does not need to continue wasting energy sending this signaling...its more energy efficient for neurons to signal/message the brain or "host/organism" ONLY when there is a change in the environmental information)**
5. Based upon his initial results, are the spikes coming from motor neurons (neurons that tell muscles to contract) or sensory neurons (neurons that send information from the periphery to the brain)? Give your reasoning. **The correct answer here must be sensory neurons because there is no longer any spine/brain/CNS (we cut leg off the body)**
6. Would stimulation of the barbs in different directions produce different spike activity? Why? **Because they are encoding different information (one carries information that a barb bent forward, and the other carries information that a barb was bent backwards)**

Garrett Brownlee
Victor Padilla
NSC 4130
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7. Assume there is a neuron that senses changes in pressure to one's skin. How would spiking activity of this sensory neuron differ from experiencing a wind gust versus a finger poke? Assume both provide the same pressure. **Since the pressure is the same, the intensity encoded within the action potential should be about/exactly the same. However, the information is encoded from different areas: wind gusts would activate the shallow mechanoreceptors (ex: Merkel cell, Meissner's corpuscle, etc.) whereas the finger poke would activate deeper mechanoreceptors (ex: Ruffini corpuscle, Pacinian corpuscle). Thus, the spiking activity would likely differ but I'm not sure how.**
8. In the space provided below, sketch your cockroach and the leg you have removed. Note any damaged parts if applicable.



Discussion Questions: Spiker Box Experiment 2

Lab 2: Changing Temperature

Instructions: Refer back to your lab manual for a description of the experiment.

1. How did the firing rate of the neurons change when placed in low temperatures? Was the change sudden, or gradual? Explain.

As the cockroach was placed into the box of dry ice, the neuronal firing rate slowly declined until coming to a complete stop. The returning of AP's (firing rate) was also gradual as well. Both of these are likely due to the time required for physiological processes to match the temperature/speed of their environment (to warm up or cool down).

2. How did the firing rate of the neurons change when placed in very high temperatures? Was the change sudden, or gradual? Explain.

The firing rate suddenly increased, but continued to suddenly increase for a while until the firing rate came to zero. Upon removing the leg away from the heat source (candle flame in our case), the firing rate never returned indicating irreversible damage was done and the neurons have died (burnt themselves out essentially)

3. At both temperature extremes, the neuron stops firing. From which temperature extreme will the neuron resume firing when returned to room temperature? Why does the neuron not resume firing when returning from the other temperature extreme?

As I previously explained (didn't realize I would be answering this), the neuron resumed firing when the leg was placed in an extreme low temperature. The inverse was true in the case of extreme high temperature (neuron did not ever resume firing), and this due to the fact that the neurons have discharged all their ions (they have essentially burnt themselves out to try and match the high energy/temperature levels of the candle flame)